

SAMIA TASMIM

Computer Science & Engineering Undergraduate

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PROFILE SUMMARY

A highly motivated and hardworking individual, I strive to hone my skills in my broad areas of interest and become a person who's well rounded in many things. I'm enthusiastic about web dev, machine learning, game dev and looking forward to learn more everyday. I've shipped projects spanning OOP simulation, game development, database-backed desktop apps, and full-stack web products with AI-assisted workflows. Passionate and interested in opportunities to contribute, collaborate with diverse teams, and tackle real world challenges in a dynamic professional setting.

Personal Information

Date Of Birth: 1-12-2006

Father's name: Dr. MD. Obaidul Islam

Mother's Name: Mrs. Israt Chowdhury

Present Address: D-13/1, Bangladesh Agricultural University Campus, Mymensingh

Permanent Address: Glorious Dream 907 A2, Mirpur, Dhaka

Home District: Bogura

EDUCATION

B.Sc. Computer Science *Islamic University Of technology* BoardBazar,Gazipur 2024-Present
(currently in 2nd year)

Higher secondary Certificarte (HSC) *Bangladesh Agricultural University College* 2023
GPA:5.00/5.00

Secondary School Certificate (SSC) *Bangladesh Agricultural University High School* 2021
GPA:5.00/5.00

SCHOLARSHIPS & AWARDS

- Awarded PSC, JSC, SSC, and HSC scholarships throughout primary and secondary education, based on merit
- Partial OIC Scholarship — awarded competitively to host-country students upon admission to IUT
- Honorable mention at junior category in national physics Olympiad

TECHNICAL SKILLS

Languages: C, C++, Java, JavaScript, TypeScript, SQL

Frameworks & Libraries: React, Spring Boot, JavaFX, raylib/raygui, Firebase, Tailwind CSS, Three.js, MediaPipe

Databases & Tooling: MySQL, Flyway, Maven, Vite, Git/GitHub, Eclipse, IntelliJ IDEA, Visual Studio Code, Codeblocks

Concepts: Object-Oriented Design, MVC & DAO patterns, RESTful APIs, Full-Stack Development, Linear Algebra, Data and telecommunication.

Real-time & Auth: Socket.IO, Firebase Auth, Google Maps API

PROJECTS

Mochi Can't Study(react-spring-app) — *full-stack study-companion web app with a reactive virtual pet (work on progress)*

Tools: React 19, TypeScript, Vite, Tailwind CSS v4, Spring Boot, MySQL, Flyway, Firebase Auth, MediaPipe, Rive

- Built a full-stack study companion where a virtual pet reacts to study sessions, focus time, and in-room interactions, with an XP/coin progression system, daily goals, and a leaderboard.
- Implemented webcam-based focus tracking with MediaPipe and an animated character state machine (Rive, with a hand-drawn SVG fallback renderer) driven by study state.
- Designed the Spring Boot + MySQL backend (Flyway migrations, Spring Security, Firebase-authenticated REST API) behind a typed, validated React/TypeScript frontend.

MovieScout — *JavaFX desktop movie browser backed by MySQL (work on progress)*

Tools: Java, JavaFX, MySQL, Eclipse, Scene Builder

- Built a Netflix-style desktop app to browse a 22-movie catalog with genre filtering, title search, cast detail pages, and a persistent Watch Later list.
- Applied an MVC + DAO architecture separating backend data access (MovieDAO, WatchLaterDAO) from FXML-driven frontend controllers.
- Implemented a live dark/light theme switch and a single-Stage view-swapping navigation shell.

Academic Apocalypse — *2D multi-room adventure game with mini-games, built on raylib*

Tools: C, raylib, raygui, custom text-to-speech module

- Developed a room-based 2D adventure in C/raylib combining a maze-navigation level, an endless-runner sequence, a graph/shortest-path puzzle, and a boss-battle mini-game.
- Built custom drag-and-drop and quiz-based mini-games (timed multiple-choice questions, ingredient-matching) with a hand-rolled GUI layer on raygui.
- Integrated a text-to-speech module and a custom animated button/menu system for game navigation.

Social Platform Simulation (oop-project) — *console-based social-media platform demonstrating core OOP design*

Tools: C++, file-based persistence, Makefile

- Modeled a Facebook-style platform (Users, Posts, Comments, Likes, Friends, Pages, Notifications, Messenger, News Feed) as a set of cooperating manager classes.
- Used a generic templated GenericManager<T> base class for shared CRUD logic across managers, applying inheritance and polymorphism throughout.
- Persisted all application state to flat text files, with an authentication service handling signup/login and session management.

Retro Love Photobooth — *browser-based retro photobooth with a Java backend*

Tools: Java (HTTP server), HTML, CSS, JavaScript (getUserMedia)

- Built a browser photobooth styled as a vintage Nikon camera, using getUserMedia for live webcam capture at up to 1080p with automatic resolution fallback.
- Implemented a 3-2-1 countdown with flash animation, in-camera photo review, retake/download options, and a lightweight Java backend to persist captures to disk.

Personal Portfolio — *personal portfolio site with WebGL-driven visual effects*

Tools: React 19, TypeScript, Vite, Three.js, OGL, postprocessing

- Designed and built a personal portfolio site with custom animated components — a WebGL galaxy background, a pixel-dissolve section reveals, and a project carousel.
- Used Three.js/OGL with a postprocessing pipeline for real-time shader effects, driven entirely from custom React components.

EXTRACURRICULAR ACTIVITIES

IUT Robotics Society

- General Member, Publications Department
- Executive Member, PR & Documentation Sub team — Project Altair

INTERESTS

Web Development, machine learning, Team work, building side projects end-to-end and AI-assisted rapid prototyping.